

Yahui Song

Software Developer, Core Strats · Standard Chartered Bank

[GitHub](#) · [Google Scholar](#) · [LinkedIn](#) · songyahui.github.io

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SUMMARY

Software engineer with a Ph.D. in program verification and 13 peer-reviewed papers, including an ACM SIGSOFT Distinguished Paper. I build **static analysis, verification and automated-repair tooling** that runs on real codebases: *ProveNFix*, my temporal-property-guided analyser and repair tool, extends Meta Infer and has been evaluated on production C. Underneath it: separation logic, temporal logic with extended regular expressions, algebraic effects and handlers, and SMT solving with Z3. I am now a software developer in Standard Chartered's Core Strats group, building trading systems in Haskell — a setting where correctness and timing carry direct financial consequence, and where I see clear potential to put these methods to work. Comfortable owning a problem end to end: theory, implementation, evaluation, and the engineering needed to get it adopted.

EDUCATION

Ph.D. in Computer Science, School of Computing, NUS 2018 – 2023

Thesis: *Symbolic Temporal Verification Techniques with Extended Regular Expressions*

Supervisor: Assoc. Prof. Wei-Ngan Chin Committee: Prof. Joxan Jaffar, Prof. Jin Song Dong

M.Comp. in Computer Science, School of Computing, NUS 2017 – 2018

Thesis: *Programming Abstraction for IoT Devices*

Supervisor: Assoc. Prof. Wei-Ngan Chin

B.Eng. in Computer Science, Sun Yat-sen University, China 2013 – 2017

Thesis: *Design and Implementation of House Roaming Interaction System Based on Unity3D*

Focus: Software Engineering, Systems Analysis, Virtual Reality

PROFESSIONAL EXPERIENCE

Software Developer — Core Strats, Standard Chartered Bank, Singapore July 2025 – Present

Building trading systems in Haskell within the bank's quantitative strategies group, where functional programming is used to keep correctness-critical financial logic tractable.

Research Fellow — Automated Program Repair Group, NUS Jan 2023 – July 2025

Worked with Prof. Abhik Roychoudhury and Prof. Wei-Ngan Chin on temporal-property-guided program repair and agentic software verification. Continued collaboration after joining SCB.

SELECTED PUBLICATIONS

8 of 13 peer-reviewed papers; full list on [Google Scholar](#). * denotes equal contribution.

TSE 2026 Detecting LLM Fact-conflicting Hallucinations Enhanced by Temporal-logic-based Reasoning

by Ningke Li*, Yahui Song*, Kailong Wang, Yuekang Li, Ling Shi, Yi Liu, Haoyu Wang
IEEE Transactions on Software Engineering

FSE 2026 Agentic Verification of Software Systems

by Haoxin Tu, Huan Zhao, Yahui Song, Mehtab Zafar, Ruijie Meng, Abhik Roychoudhury
ACM International Conference on the Foundations of Software Engineering, Montreal, Canada

SAS 2025 Specifying and Verifying Future Conditions

by Yahui Song, Darius Foo, Wei-Ngan Chin
32nd Static Analysis Symposium, Singapore

ICFP 2024 Specification and Verification for Unrestricted Algebraic Effects and Handling

by Yahui Song, Darius Foo, Wei-Ngan Chin

29th ACM SIGPLAN International Conference on Functional Programming, Milan, Italy

FM 2024 **Staged Specification Logic for Verifying Higher-Order Imperative Programs**

by Darius Foo, Yahui Song, Wei-Ngan Chin

26th International Symposium on Formal Methods, Milan, Italy

FSE 2024 **ProveNFix: Temporal Property Guided Program Repair**

by Yahui Song, Xiang Gao, Wenhua Li, Wei-Ngan Chin, Abhik Roychoudhury

ACM International Conference on the Foundations of Software Engineering, Porto de Galinhas, Brazil

ACM SIGSOFT Distinguished Paper Award

TACAS 2023 **Automated Verification for Real-Time Systems**

by Yahui Song, Wei-Ngan Chin

29th International Conference on Tools and Algorithms for the Construction and Analysis of Systems, Paris, France

VMCAI 2021 **A Synchronous Effects Logic for Temporal Verification of Pure Esterel**

by Yahui Song, Wei-Ngan Chin

22nd International Conference on Verification, Model Checking, and Abstract Interpretation

AWARDS & HONOURS

- ACM SIGSOFT Distinguished Paper Award — FSE 2024
- Honor List of Student Tutors for Excellence of Teaching, NUS, 2020–2021
- First Prize, Student Research Competition — APLAS 2018
- First Prize, GSAC 2018 Singapore Hackathon — Scheduling Algorithm Competition (Alibaba & Lazada)

SERVICE & TALKS

- **Invited talks** at ETH Zürich, EPFL, UC Santa Cruz, UNSW, and Shanghai Jiao Tong University.
- **Program committee:** JFP'26, Haskell'26, TOSEM'25, ICSE'26, ICFP'25, SRC@SPLASH'25, HOPE'25, OCaml'25, APR@ICSE'25, TASE'24 and its Special Journal Issue, etc.
- **Artifact evaluation:** CAV'23, CAV'22, PLDI'22, POPL'22, ESOP'22.
- **Teaching:** 15 semesters as a teaching assistant at NUS across seven programming languages and software engineering courses; NUS Honour List for teaching, 2020–2021.